

Topic 9: Metals

Premium Answer Booklet

23 questions	2020-2025 Paper 4	Premium readable layout
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Rebuilt for easier reading with structured tables, clear answer sections and highlighted workings. Use beside the matching topical question bank.

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Question 01

9 marks

2020 Oct/Nov Paper 41 Question 1

Main topic: Topic 9: Metals
Cross-topic links: Topic 2: Atoms, elements and compounds
Answer source label: 0620_w20_ms_41.pdf

1(a)(i)	• hydrogen / carbon
1(a)(ii)	• silicon
1(a)(iii)	• aluminium
1(a)(iv)	• iron
1(a)(v)	• aluminium
1(a)(vi)	• oxygen
1(b)(i)	• metal higher in reactivity series / metal more reactive (than iron) / allow named metal e.g. magnesium or zinc (1) • zinc corrodes/oxidises/reacts in preference to iron (1)
1(b)(ii)	• any barrier method e.g. painting

Easy explanation / reminder

- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.
- Metals tip: link the method to the reactivity series. Very reactive metals need electrolysis; less reactive metal oxides can often be
- reduced by carbon/CO.

END OF ANSWER

Question 02

9 marks

2020 Oct/Nov Paper 41 Question 2

Main topic: Topic 9: Metals
Answer source label: 0620_w20_ms_41.pdf

2(a)	• zinc blende
2(b)(i)	• $\text{ZnO} + \text{C} \rightarrow \text{Zn} + \text{CO}$ • or • $2\text{ZnO} + \text{C} \rightarrow 2\text{Zn} + \text{CO}_2$
2(b)(ii)	• chemical change: reduction (1) • explanation: oxygen is lost (1)
2(b)(iii)	• aluminium is more reactive than carbon
2(b)(iv)	• electrolysis
2(c)	• exists as layers (1) • (alloy) contains different sized (copper) atoms (1) • makes it more difficult for layers (of atoms) to slide over each slip/shift other (1)

Easy explanation / reminder

- Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.
- Metals tip: link the method to the reactivity series. Very reactive metals need electrolysis; less reactive metal oxides can often be
- reduced by carbon/CO.

END OF ANSWER

Question 03

12 marks

2020 Oct/Nov Paper 42 Question 3

Main topic: Topic 9: Metals
Cross-topic links: Topic 8: The Periodic Table
Answer source label: 0620_w20_ms_42.pdf

3(a)	• any two from: • shiny / lustrous • conduct electricity • conduct heat
3(b)	• low(er) density (1) • low(er) melting points (1)
3(c)(i)	• hydrogen
3(c)(ii)	• hydroxide / OH⁻
3(c)(iii)	• $2\text{Na} + 2\text{H}_2\text{O} \rightarrow 2\text{NaOH} + \text{H}_2$ • NaOH (as a product) (1) • rest of equation (1)
3(d)(i)	• galvanising
3(d)(ii)	• prevents water or / and oxygen reaching iron
3(d)(iii)	• zinc more reactive (than iron) (1) • zinc corrodes / oxidises / reacts in preference to iron (1)

Easy explanation / reminder

- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.
- Metals tip: link the method to the reactivity series. Very reactive metals need electrolysis; less reactive metal oxides can often be
- reduced by carbon/CO.

END OF ANSWER

Question 04

6 marks

2020 Oct/Nov Paper 43 Question 1

Main topic: Topic 9: Metals

Cross-topic links: Topic 11: Organic chemistry, Topic 7: Acids, bases and salts, Topic 10: Chemistry of the environment, Topic 4: Electrochemistry

Answer source label: 0620_w20_ms_43.pdf

1(a)	• iron(III) oxide
1(b)	• nitrogen dioxide
1(c)	• ammonia
1(d)	• silver
1(e)	• carbon monoxide
1(f)	• anhydrous cobalt(II) chloride

Easy explanation / reminder

- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.
- Organic tip: identify the functional group first. Combustion of hydrocarbons makes CO₂ and H₂O when oxygen is in excess.

END OF ANSWER

Question 05

17 marks

2020 Oct/Nov Paper 43 Question 4

Main topic: Topic 9: Metals
Cross-topic links: Topic 3: Stoichiometry
Answer source label: 0620_w20_ms_43.pdf

4(a)	• sulfur dioxide
4(b)(i)	• Any two from • (coke) • releases heat or releases energy(when it reacts with oxygen or burns in air) / (acts as a) fuel / increases temperature (in the furnace) / heats (the furnace) / source of energy • (coke) • reduces zinc oxide / is a reducing agent / converts zinc oxide to zinc / removes oxygen from zinc oxide • (coke) • (reacts with oxygen) to produce carbon monoxide / reacts with carbon dioxide to form carbon monoxide • carbon monoxide reduces zinc oxide / converts zinc oxide to zinc / removes oxygen from zinc oxide
4(b)(ii)	• $\text{ZnO} + \text{CO} \rightarrow \text{Zn} + \text{CO}_2$ • OR • $2\text{ZnO} + \text{C} \rightarrow 2\text{Zn} + \text{CO}_2$ • OR • $\text{ZnO} + \text{C} \rightarrow \text{Zn} + \text{CO}$
4(b)(iii)	• temperature inside furnace is above / higher than 907 (°C) • OR temperature is above / higher than the boiling point (of zinc) opposite reasoning also allowed • OR 1200 (°C) is above / higher than the boiling point (of zinc) opposite reasoning also allowed • OR 1200 (°C) is above / higher than 907 (°C) opposite reasoning also allowed
4(b)(iv)	• condensation / condensing
4(c)(i)	• zinc
4(c)(ii)	• (a solution containing the) maximum amount of solute dissolved / no more solute can dissolve (1) • at a given temperature (1)
4(c)(iii)	• one mark for each of any two from: • zinc oxide • zinc hydroxide • zinc carbonate
4(d)(i)	• heat again and weigh again / repeat steps 2 and 3 (1) • until mass is constant (1)
4(d)(ii)	• (moles of MgSO_4 =) 0.005 / 5×10^{-3} (1) • mass of water = 0.63 g (1) • moles of water = $0.63 / 18 = 0.035$ / 3.5×10^{-2} (1) • ($x = 0.035 / 0.005$) = 7 (1)

Easy explanation / reminder

- Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
- before converting to the final answer.
- Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.

END OF ANSWER

Question 06

13 marks

2021 May/June Paper 41 Question 6

Main topic: Topic 9: Metals
Answer source label: 0620_s21_ms_41.pdf

6(a)(i)	• hematite
6(a)(ii)	• air
6(a)(iii)	• slag / calcium silicate
6(a)(iv)	• any two from: • (coke) • releases heat (when it reacts with oxygen or reacts in air) • OR (acts as a) fuel • OR increases temperature (in the furnace) / heats (the furnace) • OR source of energy • (coke or carbon monoxide) • reduces iron oxide • OR is a reducing agent • OR converts iron oxide to iron / • removes oxygen from iron oxide • (coke) • reacts with oxygen to form carbon monoxide • OR reacts with carbon dioxide to form carbon monoxide
6(b)	• S22- or S-
6(c)(i)	• any two from: • (iron forms) coloured compounds • (iron has) variable oxidation states • (iron acts as a) catalyst
6(c)(ii)	• any two from: • (iron is) good conductor of electricity • (iron) forms a basic oxide • (iron salts are) soluble (in water)
6(d)(i)	• magnesium is more reactive than iron / steel opposite reasoning also allowed (1) • iron is not oxidised • OR • iron does not lose electrons • OR • magnesium loses electrons more easily than or in preference (to iron) opposite reasoning also allowed • OR • magnesium is oxidised more easily or reacts with oxygen more easily or corrodes more easily or in preference (to iron) • opposite reasoning also allowed (1)
6(d)(ii)	• copper is less reactive than iron / copper is lower in the reactivity series than iron opposite reasoning also allowed

Easy explanation / reminder

- Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
- before converting to the final answer.
- Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than
- activation energy.

END OF ANSWER

Question 07

13 marks

2021 May/June Paper 43 Question 6

Main topic: Topic 9: Metals
Answer source label: 0620_s21_ms_43.pdf

6(a)(i)	• limestone
6(a)(ii)	• (molten) iron
6(a)(iii)	• coke / carbon / carbon monoxide
6(a)(iv)	• (reaction between) coke / carbon and oxygen • Question Answer
6(a)(v)	• nitrogen (1) • carbon dioxide (1)
6(b)(i)	• zinc sulfide
6(b)(ii)	• condensation/condensing
6(c)	• brass
6(d)(i)	• any two from (copper has): • high density • high melting point • variable oxidation states
6(d)(ii)	• any two from (copper): • good conductor of electricity • forms a basic oxide • soluble salts • Question Answer

Easy explanation / reminder

- Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.
- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

END OF ANSWER

Question 08

18 marks

2021 Oct/Nov Paper 41 Question 1

Main topic: Topic 9: Metals
Answer source label: 0620_w21_ms_41.pdf

1(a)(i)	• hydrogen
1(a)(ii)	• sodium
1(a)(iii)	• iron
1(a)(iv)	• aluminium
1(a)(v)	• zinc or copper
1(a)(vi)	• iron
1(b)(i)	• zinc blende
1(b)(ii)	• bauxite
1(c)	• protective oxide layer
1(d)	• Any 2 from 3 • zinc • iron • copper
1(e)(i)	• copper
1(e)(ii)	• zinc sulfate
1(e)(iii)	• particles have more energy • More collisions (between particles) occur per second / per unit time • greater percentage / proportion / fraction of collisions (of particles) are successful / have energy above activation • energy / have energy equal to activation energy
1(e)(iv)	• Any 2 from 3: • use a catalyst • use smaller granules • increase concentration

Easy explanation / reminder

- Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
- before converting to the final answer.
- Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than
- activation energy.

END OF ANSWER

Question 09

9 marks

2022 May/June Paper 41 Question 1

Main topic: Topic 9: Metals
Cross-topic links: Topic 11: Organic chemistry
Answer source label: 0620_s22_ms_41.pdf

1(a)	• carbon dioxide
1(b)	• aluminium oxide
1(c)	• ethanol (1) • carbon dioxide (1)
1(d)	• ethanol
1(e)	• oxygen
1(f)	• carbon dioxide
1(g)	• nitrogen
1(h)	• diamond

Easy explanation / reminder

- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.
- Metals tip: link the method to the reactivity series. Very reactive metals need electrolysis; less reactive metal oxides can often be
- reduced by carbon/CO.

END OF ANSWER

Question 10

8 marks

2022 May/June Paper 42 Question 1

Main topic: Topic 9: Metals

Cross-topic links: Topic 10: Chemistry of the environment, Topic 8: The Periodic Table

Answer source label: 0620_s22_ms_42.pdf

1(a)	• Mg
1(b)	• Ar
1(c)	• Cl
1(d)	• Si
1(e)	• P
1(f)	• Na
1(g)	• Al
1(h)	• S

Easy explanation / reminder

- Metals tip: link the method to the reactivity series. Very reactive metals need electrolysis; less reactive metal oxides can often be
- reduced by carbon/CO.

END OF ANSWER

Question 11

6 marks

2023 May/June Paper 42 Question 1

Main topic: Topic 9: Metals
Cross-topic links: Topic 10: Chemistry of the environment
Answer source label: 0620_s23_ms_42.pdf

1(a)	• D
1(b)	• C
1(c)	• H
1(d)	• B
1(e)	• C
1(f)	• G

Easy explanation / reminder

- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.
- Metals tip: link the method to the reactivity series. Very reactive metals need electrolysis; less reactive metal oxides can often be
- reduced by carbon/CO.

END OF ANSWER

Question 12

12 marks

2023 Oct/Nov Paper 41 Question 3

Main topic: Topic 9: Metals
Cross-topic links: Topic 8: The Periodic Table
Answer source label: 0620_w23_ms_41.pdf

3(a)(i)	• Ag column all $\times$ (1) • $\times$ in Pb AND 2 $\checkmark$ in Zn (1) • Zn, Mn, Pb Ag (1)
3(a)(ii)	• (all) nitrates are soluble OR lead sulfate is insoluble
3(a)(iii)	• $\text{Zn} + 2\text{AgNO}_3 \rightarrow \text{Zn}(\text{NO}_3)_2 + 2\text{Ag}$ • $\text{Zn}(\text{NO}_3)_2$ on the right hand side (1) • correct equation (1)
3(b)(i)	• colourless (1) • orange (1)
3(b)(ii)	• $\text{Cl}_2(\text{g}) + 2\text{Br}^-(\text{aq}) \rightarrow \text{Br}_2(\text{aq}) + 2\text{Cl}^-(\text{aq})$ • $\text{Br}_2 + \text{Cl}^-$ as products (1) • correct equation (1) • state symbols (1)
3(b)(iii)	• tenessine / Ts

END OF ANSWER

Question 13

17 marks

2023 Oct/Nov Paper 42 Question 3

Main topic: Topic 9: Metals
Answer source label: 0620_w23_ms_42.pdf

3(a)	• (hot) air
3(b)(i)	• hematite
3(b)(ii)	• $\text{Fe}_2\text{O}_3 + 3\text{CO} \rightarrow 3\text{CO}_2 + 2\text{Fe}$
3(b)(iii)	• from +3 (1) • to 0 (1)
3(b)(iv)	• decrease (in oxidation number)
3(c)	• $\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$ (1) • thermal decomposition (1) • $\text{SiO}_2 + \text{CaO} \rightarrow \text{CaSiO}_3$ (1) • acid-base reaction (1)
3(d)(i)	• alloy
3(d)(ii)	• nickel / chromium
3(e)	• hydrated iron(III) oxide
3(f)(i)	• galvanising
3(f)(ii)	• painting / greasing / coating with plastic
3(f)(iii)	• zinc more reactive (than iron) (1) • zinc corrodes / oxidises / reacts in preference to / instead of iron (1)

Easy explanation / reminder

- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.
- Metals tip: link the method to the reactivity series. Very reactive metals need electrolysis; less reactive metal oxides can often be
- reduced by carbon/CO.

END OF ANSWER

Question 14

18 marks

2023 Oct/Nov Paper 43 Question 5

Main topic: Topic 9: Metals
Answer source label: 0620_w23_ms_43.pdf

5(a)(i)	• positive ions / cations (1) • sea of electrons / mobile electrons / delocalised electrons (1) • attraction between positive ions and electrons (1)
5(a)(ii)	• electrons move / electrons mobile / electrons flow
5(b)(i)	• any two • (coke) releases heat • (coke) reduces iron(III) oxide • (coke) reacts with carbon dioxide to form carbon monoxide
5(b)(ii)	• limestone decomposes to calcium oxide (1) • calcium oxide reacts with / neutralises silicon(IV) oxide to produce slag / calcium silicate (1)
5(b)(iii)	• $\text{Fe}_2\text{O}_3 + 3\text{C} \rightarrow 2\text{Fe} + 3\text{CO}$ • OR • $\text{Fe}_2\text{O}_3 + 3\text{CO} \rightarrow 2\text{Fe} + 3\text{CO}_2$ • OR • $2\text{Fe}_2\text{O}_3 + 3\text{C} \rightarrow 4\text{Fe} + 3\text{CO}_2$ • all formulae (1) • equation correct (1)
5(b)(iv)	• temperature in the furnace is above the melting point of iron
5(c)	• atoms have different sizes (1) • layers cannot slide over or past each other (1)
5(d)(i)	• oxygen AND water
5(d)(ii)	• galvanising
5(d)(iii)	• acts as a barrier which prevents oxygen and water reaching the iron
5(d)(iv)	• zinc is more reactive than iron (1) • zinc is oxidised / zinc loses electrons / zinc forms positive ions / forms zinc ions (1)

Easy explanation / reminder

- Calculation tip: start with $\text{moles} = \text{mass} / \text{Mr}$ or $\text{moles} = \text{concentration} \times \text{volume}$. Then use the mole ratio from the balanced equation
- before converting to the final answer.
- Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.

END OF ANSWER

Question 15

19 marks

2024 Feb/March Paper 42 Question 1

Main topic: Topic 9: Metals
Answer source label: 0620_m24_ms_42.pdf

1(a)	• coke
1(a)(ii)	• provide heat
1(b)(i)	• 160 • 112 AND 70.(0)(%)
1(b)(ii)	• hematite
1(b)(iii)	• by reduction of carbon dioxide
1(b)(iv)	• $\text{Fe}_2\text{O}_3 + 3\text{CO} \rightarrow 2\text{Fe} + 3\text{CO}_2$ • species • correct equation
1(b)(v)	• reduction
1(c)	• thermal decomposition
1(d)(i)	• CaO is basic • SiO₂ is acidic
1(d)(ii)	• silicon(IV) oxide • slag
1(e)(i)	• aluminium is above carbon in the reactivity series • OR • aluminium is more reactive than carbon • February/March 2024
1(e)(ii)	• electrolysis
1(f)(i)	• 2,8
1(f)(ii)	• 26 (protons) • 23 (electrons)

Easy explanation / reminder

- Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.
- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

END OF ANSWER

Question 16

14 marks

2024 May/June Paper 41 Question 6

Main topic: Topic 9: Metals
Answer source label: 0620_s24_ms_41.pdf

6(a)(i)	• limestone • OR • calcium carbonate
6(a)(ii)	• (molten) iron
6(a)(iii)	• silicon(IV) oxide • OR • silicon dioxide
6(a)(iv)	• coke or carbon and oxygen
6(a)(v)	• Any two from: • $\blacksquare$ • nitrogen • $\blacksquare$ • carbon dioxide • $\blacksquare$ • argon
6(b)(i)	• the temperature in the furnace is above or higher than the boiling point of zinc opposite reasoning also allowed • OR • the boiling point of zinc is below or less than the temperature of the furnace opposite reasoning also allowed
6(b)(ii)	• condensation / condensing
6(c)(i)	• galvanising
6(c)(ii)	• zinc is more reactive than iron (1) • zinc is oxidised / • zinc loses electrons / • zinc forms positive ions / • zinc forms zinc ions (1)
6(d)(i)	• amphoteric
6(d)(ii)	• Na_2ZnO_2
6(d)(iii)	• zinc sulfate

Easy explanation / reminder

- Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.
- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

END OF ANSWER

Question 17

14 marks

2024 May/June Paper 42 Question 2

Main topic: Topic 9: Metals
Cross-topic links: Topic 7: Acids, bases and salts
Answer source label: 0620_s24_ms_42.pdf

2(a)(i)	• electron loss (from outer shell)
2(a)(ii)	• potassium
2(a)(iii)	• magnesium or potassium
2(a)(iv)	• iron
2(b)	• Colourless • Blue • Colour given in
2(c)(i)	• hydrogen
2(c)(ii)	• strontium hydroxide
2(c)(iii)	• magnesium or Mg • magnesium oxide
2(d)	• $\text{Fe}_3\text{O}_4 + 8\text{HCl} \rightarrow 2\text{FeCl}_3 + \text{FeCl}_2 + 4\text{H}_2\text{O}$ • FeCl_2 or FeCl_3 as product of equation • FeCl_2 and FeCl_3 and H_2O as only products of equation • Correct equation

Easy explanation / reminder

- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

END OF ANSWER

Question 18

13 marks

2024 May/June Paper 43 Question 6

Main topic: Topic 9: Metals
Answer source label: 0620_s24_ms_43.pdf

6(a)(i)	• hematite
6(a)(ii)	• air
6(a)(iii)	• carbon monoxide
6(a)(iv)	• limestone thermally decomposes (into calcium oxide and carbon dioxide) • calcium oxide reacts with silicon(IV) oxide • producing slag
6(b)(i)	• have low densities • or • have low melting point
6(b)(ii)	• act as catalysts • or • form coloured compounds
6(b)(iii)	• form basic oxides • or • form positive ions
6(b)(iv)	• form acidic oxides • or • form negative ions
6(c)(i)	• magnesium loses electrons more readily than / in preference to / instead of iron • magnesium is more reactive than iron opposite reasoning also allowed
6(c)(ii)	• any metal below iron in the reactivity series e.g. copper / silver / gold

Easy explanation / reminder

- Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than
- activation energy.
- Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.

END OF ANSWER

Question 19

9 marks

2024 Oct/Nov Paper 43 Question 1

Main topic: Topic 9: Metals

Cross-topic links: Topic 2: Atoms, elements and compounds, Topic 11: Organic chemistry, Topic 4: Electrochemistry, Topic 7: Acids, bases and

Answer source label: 0620_w24_ms_43.pdf

1(a)	• ethanol / carbon dioxide
1(b)	• helium
1(c)	• carbon dioxide
1(d)	• cryolite
1(e)	• hematite
1(f)	• hydrogen
1(g)	• silicon(IV) oxide
1(h)	• graphite
1(i)	• carbon dioxide

Easy explanation / reminder

- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.
- Metals tip: link the method to the reactivity series. Very reactive metals need electrolysis; less reactive metal oxides can often be
- reduced by carbon/CO.

END OF ANSWER

Question 20

15 marks

2025 May/June Paper 41 Question 3

Main topic: Topic 9: Metals
Cross-topic links: Topic 7: Acids, bases and salts
Answer source label: 0620_s25_ms_41.pdf

3(a)(i)	• positive ions / cations(1) • sea of electrons / • mobile electrons / • delocalised electrons(1) • attraction between positive ions and electrons(1)
3(a)(ii)	• electrons move / • electrons mobile / • electrons flow
3(b)(i)	• harder or stronger
3(b)(ii)	• zinc
3(c)(i)	• Any two from:- • solid disappears / solid dissolves • fizzing / bubbling / effervescence • blue solution
3(c)(ii)	• all the sulfuric acid had reacted
3(c)(iii)	• copper(II) carbonate
3(c)(iv)	• copper(II) oxide / copper(II) hydroxide
3(c)(v)	• saturated (solution)
3(c)(vi)	• more particles per unit volume • more collisions per unit time / increased collision frequency
3(c)(vii)	• hydrated

Easy explanation / reminder

- Calculation tip: start with $\text{moles} = \text{mass} / M_r$ or $\text{moles} = \text{concentration} \times \text{volume}$. Then use the mole ratio from the balanced equation
- before converting to the final answer.
- Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than
- activation energy.

END OF ANSWER

Question 21

9 marks

2025 May/June Paper 42 Question 4

Main topic: Topic 9: Metals
Cross-topic links: Topic 3: Stoichiometry
Answer source label: 0620_s25_ms_42.pdf

4(a)(i)	• 4CO
4(a)(ii)	• chromium will need to be separated from iron
4(b)	• alloy
4(c)(i)	• -2
4(c)(ii)	• use of -14 or 7 -> -2 • +6
4(c)(iii)	• mol $(\text{NH}_4)_2\text{Cr}_2\text{O}_7 = 1.26 / 252 = 0.005(00)$ • mol of $\text{N}_2 = 0.005(00)$ • vol of $\text{N}_2 = 0.005 \rightarrow 24000 = 120 \text{ (cm}^3\text{)}$

Easy explanation / reminder

- Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
- before converting to the final answer.
- Metals tip: link the method to the reactivity series. Very reactive metals need electrolysis; less reactive metal oxides can often be
- reduced by carbon/CO.

END OF ANSWER

Question 22

15 marks

2025 May/June Paper 43 Question 3

Main topic: Topic 9: Metals

Cross-topic links: Topic 7: Acids, bases and salts, Topic 2: Atoms, elements and compounds

Answer source label: 0620_s25_ms_43.pdf

3(a)(i)	• metallic (bonding)
3(a)(ii)	• positive ions • electrons
3(a)(iii)	• electrons
3(b)(i)	• a mixture of a metal with other elements
3(b)(ii)	• copper
3(c)(i)	• solid stops dissolving / disappearing • no more bubbles / fizzing / effervescence on addition of solid
3(c)(ii)	• aqueous zinc sulfate
3(c)(iii)	• zinc oxide • OR • zinc hydroxide
3(c)(iv)	• (a solution containing the) maximum concentration of a solute dissolved in the solvent • at a specified temperature
3(c)(v)	• (large pieces have) smaller surface area • frequency of collisions between (zinc carbonate and acid) particles decreases
3(c)(vi)	• (crystals that are) chemically combined with water

Easy explanation / reminder

- Calculation tip: start with $\text{moles} = \text{mass} / M_r$ or $\text{moles} = \text{concentration} \times \text{volume}$. Then use the mole ratio from the balanced equation
- before converting to the final answer.
- Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than
- activation energy.

END OF ANSWER

Question 23

19 marks

2025 Oct/Nov Paper 42 Question 2

Main topic: Topic 9: Metals

Cross-topic links: Topic 2: Atoms, elements and compounds, Topic 6: Chemical reactions

Answer source label: 0620_w25_ms_42.pdf

2(a)(i)	• hematite
2(a)(ii)	• burning of carbon (coke)
2(a)(iii)	• carbon monoxide
2(a)(iv)	• $3\text{CO} + \text{Fe}_2\text{O}_3 \rightarrow 2\text{Fe} + 3\text{CO}_2$ • species • correct equation
2(a)(v)	• $\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$ • $\text{CaO} + \text{SiO}_2 \rightarrow \text{CaSiO}_3$
2(b)	• atoms of the same element that have the same number of protons • but different numbers of neutrons
2(c)(i)	• $p = 26$ and 26 • $n = 28$ and 32 • $e = 24$ and 23
2(c)(ii)	• NaOH(aq) OR $\text{NH}_3\text{(aq)}$ • green ppt. • (insoluble in excess, ppt. turns brown near surface on standing)
2(d)(i)	• a substance that oxidises another substance and is itself reduced • 'a substance' oxidises another substance • is itself reduced
2(d)(ii)	• loss of (one) electron
2(d)(iii)	• acidified
2(e)	• iodine

Easy explanation / reminder

- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.
- Metals tip: link the method to the reactivity series. Very reactive metals need electrolysis; less reactive metal oxides can often be
- reduced by carbon/CO.

END OF ANSWER